21. Calculate the seasonal indices from the following data using the average from the following data using the average method:

	I Quarter	II Quarter	III Quarter	IV Quarter
2008	72	68	62	76
2009	78	74	78	72
2010	74	70	72	76
2011	76	74	74	72
2012	72	72	76	68

22. The following table shows the number of salesmen working for a certain concern:

Year	1992	1993	1994	1995	1996
No. of salesmen	46	48	42	56	52

Use the method of least squares to fit a straight line and estimate the number of salesmen in 1997.

9.2 Index Number

Introduction:

Index Numbers are the indicators which reflect the changes over a specified period of time in price of different commodities, production, sales, cost of living etc... Index Numbers are statistical methods used to measure the relative change in the level of a variable or group of variables with respect to time, geographical location or other characteristics such as income, profession etc. The variables may be

(i) The price of a particular commodity. For example gold, silver, iron (or) a group of commodities. For example consumer goods, household food items etc..

(ii) The volume of export and import, agricultural and industrial production.

(iii) National income of a country, cost of living of persons belonging to a particular income group.

9.2.1 Meaning, Classifications and Uses

Suppose we want to measure the general changes in the price level of consumer goods, the price changes are not directly measurable, as the prices of various commodities are in different units For example rice, wheat and sugar are in kilograms, milk, petrol, oil are in litres, clothes are in metres etc… Further, the price and quantity of some commodities may increase or decrease during the two time periods. Therefore, index number gives a single representative value which gives the general level of the prices of the commodities in a given group over a specified time period.

Definition 9.2

"An Index Number is a device which shows by its variations the Changes in a magnitude which is not capable of accurate measurements in itself or of direct valuation in practice". - *Wheldon*

"An Index number is a statistical measure of fluctuations in a variable arranged in the form of a series and using a base period for making comparisons" – *Lawrence J Kalpan*

Classification of Index Numbers:

Index number can be classified as follows,

(i) Price Index Number

It measures the general changes in the retail or wholesale price level of a particular or group of commodities.

(ii) Quantity Index Number

These are indices to measure the changes in the quantity of goods manufactured in a factory.

(iii) Cost of living Index Number

These are intended to study the effect of change in the price level on the cost of living of different classes of people.

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